Name: _____

TOPIC TEST 1

Chapter 1 Derivatives, exponential and trigonometric functions

- Time allowed: 70 minutes
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (58 marks).
- Part C: Two analysis questions (15 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which statement is *not* true?

A $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

B $\frac{d}{dx}(e^{f(x)}) = e^{f(x)}$

C $\frac{d}{dx}[\tan(x)] = \frac{1}{\cos^2(x)}$

D $\frac{d}{dx}(uv) = vu' + uv'$

E $\frac{d}{dx}\{\cos[f(x)]\} = -f'(x)\sin[f(x)]$

[1 mark]

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2 The gradient of the tangent to the curve $y = \cos(2x)$ at the point where $x = \frac{\pi}{12}$ is

A $-\frac{1}{2}$

B 1

C -1

D $\sqrt{3}$

E $-\sqrt{3}$

[1 mark]

3 Which is the correct derivative of $\frac{2x+5}{6x-1}$?

A $\frac{28}{(6x-1)^2}$

B $\frac{-32}{(6x-1)^2}$

C $\frac{32}{(6x-1)^2}$

D $\frac{-28}{(6x-1)^2}$

E $\frac{-7}{(6x-1)^2}$

[1 mark]

4 The initial velocity of a particle with displacement $x = 4e^{3t} + 2t^2 - 3t + 5$ is:

A 1

B 12

C 13

D 9

E 4

[1 mark]

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5 Given $Q = Q_0e^{kt}$, which is *not* a correct statement?

A The equation shows that Q is growing exponentially over time t.

B $\frac{dQ}{dt} = kQ$

C $\frac{dQ}{dt} = kQ_0e^{kt}$

D $\frac{dQ}{dt} = ke^{kt}$

E $\frac{dQ}{dt}$ is the rate at which Q is changing over time t.

[1 mark]

Part B

Short-answer questions

Six short-answer questions
58 marks
Show your working where appropriate.

1 Differentiate:

a x^{-9}

b $\frac{3}{x^3}$

c $\sqrt[4]{x}$

d $x + \frac{1}{x}$

e $e^x + \tan(x)$

[9 marks]

2 Find the derivative of:

a $2x(3x - 4)^5$

b $9(x^3 + 1)^3$

c $\dfrac{5x + 2}{x^2}$

d $[1 + \sin(2x)]^7$

e $\dfrac{e^x}{x}$

[12 marks]

3 Find the gradient of the tangent to each curve:

a $y = \sqrt{3x + 1}$ at $x = 5$

b $y = \cos\left(\frac{x}{3}\right)$ at $x = \pi$

c $y = x + e^{2x}$ at $x = 3.9$ (answer correct to 1 decimal place).

[8 marks]

4 Find the equation of the tangent to each curve:

a $y = 2\sin(3x)$ at the point $(\pi, 0)$

b $y = e^{4x}$ at the point $(2, e^8)$

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c $y = \tan(x)$ at the point where $x = \frac{\pi}{4}$.

[13 marks]

5 A particle moves in simple harmonic motion so that its displacement, x cm, over time, t seconds, is given by $x = 3\cos(\pi t)$. Find:

a the displacement after 7 seconds

b the velocity after $\frac{1}{2}$ second

c the initial acceleration

d the times when the particle is in equilibrium

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e the acceleration when the particle is at rest.

[11 marks]

6 A long-term project to help people in Beachtown gain employment uses a model of $N = 2400e^{-0.11t}$ for the number of unemployed people over t years.

a How many people were unemployed at the start of the project?

b How many people were unemployed after 6 years?

c What was the unemployment rate:

i initially?

ii after 6 years?

[5 mark]

Part C

Analysis questions

Two analysis questions
15 marks
Show your working where appropriate.

1 The volume, V kL, of water in a dam over time, t weeks, is given by

$V = 5000 + 1300e^{0.034t}$

a What was the initial volume in the dam, correct to 1 decimal place?

b Find the volume, correct to 1 decimal place, after

i 10 weeks

ii 6 months.

c Find the rate at which the volume, correct to 1 decimal place, is changing:

i initially

ii after10 weeks

iii after 6 months.

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d Show that $\frac{dV}{dt} = 0.034(V - 5000)$.

[9 marks]

2 A pendulum moves 4.5 cm to either side of its centre and it takes 0.2 seconds to travel from the centre out to one side and back into the centre.

a Write an equation in the form $x = a \sin(bt)$ for the movement of the pendulum.

b Find the maximum speed at which the pendulum moves.

[6 marks]

[Total marks: 78]

Answers

Part A – Multiple-choice questions [1 mark each]

1 The statement $\frac{d}{dx}(e^{f(x)}) = e^{f(x)}$ is not correct since $\frac{d}{dx}(e^{f(x)}) = f'(x)e^{f(x)}$.

$\therefore$ **B**

2 When $x = \frac{\pi}{12}$

$$\begin{aligned}\frac{dy}{dx} &= -2\sin(2x)\\ &= -2\sin\left(\frac{\pi}{6}\right)\\ &= -2 \times \frac{1}{2}\\ &= -1\end{aligned}$$

$\therefore$ **C**

3 $$\frac{dy}{dx} = \frac{(6x-1)\times 2 - (2x+5)\times 6}{(6x-1)^2}$$

$$= \frac{-32}{(6x-1)^2}$$

$\therefore$ **B**

4 When $t = 0$

$$\begin{aligned}v &= 12e^{3t} + 4t - 3\\ &= 12e^0 + 0 - 3\\ &= 12 - 3\\ &= 9\end{aligned}$$

$\therefore$ **D**

5 Since $\frac{dQ}{dt} = kQ_0e^{kt}$, the statement $\frac{dQ}{dt} = ke^{kt}$ is wrong.

$\therefore$ **D**

Part B – Short-answer questions

1 a $$\begin{aligned}\frac{d}{dx}(x^{-9}) &= -9x^{-9-1}\\ &= -9x^{-10}\end{aligned}$$ **[1 mark]**

b $$\frac{3}{x^3} = 3x^{-3}$$

$$\frac{d}{dx}(3x^{-3}) = -9x^{-4}$$ **[1 mark]**

$$= -\frac{9}{x^4}$$ **[1 mark]**

c $\sqrt[4]{x} = x^{\frac{1}{4}}$

$$\frac{d}{dx}(x^{\frac{1}{4}}) = \frac{1}{4}x^{\frac{1}{4}-1}$$

$= \frac{1}{4}x^{-\frac{3}{4}}$ **[1 mark]**

$= \frac{1}{4\sqrt[4]{x^3}}$ **[1 mark]**

d $x + \frac{1}{x} = x + x^{-1}$

$\frac{d}{dx}(x + x^{-1}) = 1 - x^{-2}$ **[1 mark]**

$= 1 - \frac{1}{x^2}$ **[1 mark]**

e $\frac{d}{dx}[e^x + \tan(x)] = e^x + \frac{1}{\cos^2(x)}$ **[2 marks]**

2 a $\frac{d}{dx}[2x(3x-4)^5] = (3x-4)^5 \times 2 + 2x \times [3 \times 5(3x-4)^4]$ **[1 mark]**

$= 2(3x-4)^5 + 30x(3x-4)^4$ **[1 mark]**

$= 2(3x-4)^4[3x - 4 + 15x]$

$= 2(3x-4)^4(18x-4)$

$= 4(3x-4)^4(9x-2)$ **[1 mark]**

b $\frac{d}{dx}[9(x^3+1)^3] = 9 \times 3x^2 \times 3(x^3+1)^2$ **[1 mark]**

$= 81x^2(x^3+1)^2$ **[1 mark]**

c $\frac{d}{dx}\left(\frac{5x+2}{x^2}\right) = \frac{x^2 \times 5 - (5x+2) \times 2x}{(x^2)^2}$ **[1 mark]**

$= \frac{5x^2 - 10x^2 - 4x}{x^4}$ **[1 mark]**

$= \frac{-5x^2 - 4x}{x^4}$

$= -\frac{5x+4}{x^3}$ **[1 mark]**

d $\frac{d}{dx}[1 + \sin(2x)]^7 = 2\cos(2x) \times 7[1 + \sin(2x)]^6$ **[1 mark]**

$= 14\cos(2x)[1 + \sin(2x)]^6$ **[1 mark]**

e $\frac{d}{dx}\left(\frac{e^x}{x}\right) = \frac{x \times e^x - e^x \times 1}{x^2}$ **[1 mark]**

$= \frac{xe^x - e^x}{x^2}$

$= \frac{e^x(x-1)}{x^2}$ **[1 mark]**

3 a $y = \sqrt{3x+1}$

$= (3x+1)^{\frac{1}{2}}$

$\frac{d}{dx}(3x+1)^{\frac{1}{2}} = 3 \times \frac{1}{2}(3x+1)^{-\frac{1}{2}}$ **[1 mark]**

$= \frac{3}{2\sqrt{3x+1}}$

When $x = 5$, $\frac{dy}{dx} = \frac{3}{2\sqrt{3 \times 5 + 1}}$ **[1 mark]**

$= \frac{3}{2\sqrt{16}}$

$= \frac{3}{8}$ **[1 mark]**

b $\frac{d}{dx}\left[\cos\left(\frac{x}{3}\right)\right] = -\frac{1}{3}\sin\left(\frac{x}{3}\right)$ **[1 mark]**

When $x = \pi$, $\frac{dy}{dx} = -\frac{1}{3}\sin\left(\frac{\pi}{3}\right)$

$= -\frac{1}{3} \times \frac{\sqrt{3}}{2}$ **[1 mark]**

$= -\frac{\sqrt{3}}{6}$ **[1 mark]**

c $\frac{d}{dx}(x + e^{2x}) = 1 + 2e^{2x}$ **[1 mark]**

When $x = 3.9$, $\frac{dy}{dx} = 1 + 2e^{7.8}$

$= 4882.2$ **[1 mark]**

4 a $\frac{d}{dx}[2\sin(3x)] = 6\cos(3x)$ **[1 mark]**

At $(\pi, 0)$, $\frac{dy}{dx} = 6\cos(3\pi)$

$= -6$ **[1 mark]**

$y - y_1 = m(x - x_1)$

$y - 0 = -6(x - \pi)$ **[1 mark]**

$y = -6x + 6\pi$ **[1 mark]**

b $\frac{d}{dx}(e^{4x}) = 4e^{4x}$ **[1 mark]**

At $(2, e^8)$, $\frac{dy}{dx} = 4e^8$ **[1 mark]**

$y - e^8 = 4e^8(x - 2)$ **[1 mark]**

$= 4e^8x - 8e^8$

$y = 4e^8x - 7e^8$ **[1 mark]**

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c $\frac{d}{dx}[\tan(x)] = \frac{1}{\cos^2(x)}$ **[1 mark]**

When $x = \frac{\pi}{4}, y = \tan\left(\frac{\pi}{4}\right) = 1$ **[1 mark]**

When $x = \frac{\pi}{4}, \frac{dy}{dx} = \frac{1}{\cos^2\left(\frac{\pi}{4}\right)}$

$= \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2}$

$= \frac{1}{\frac{1}{2}}$

$= 2$ **[1 mark]**

$y - 1 = 2\left(x - \frac{\pi}{4}\right)$ **[1 mark]**

$= 2x - \frac{\pi}{2}$

$2y - 2 = 4x - \pi$

$4x - 2y + 2 - \pi = 0$ **[1 mark]**

5 a When $t = 7, x = 3\cos(7\pi)$

$= -3$ cm **[1 mark]**

b $v = -3\pi \sin(\pi t)$ **[1 mark]**

When $t = \frac{1}{2}, v = -3\pi \sin\left(\frac{\pi}{2}\right)$

$= -3\pi$ cm/s **[1 mark]**

c $a = -3\pi^2 \cos(\pi t)$ **[1 mark]**

When $t = 0$, a $= -3\pi^2 \cos(0)$

$= -3\pi^2$ cm/s^2 **[1 mark]**

d At equilibrium, $x = 0$

$0 = 3\cos(\pi t)$ **[1 mark]**

$0 = \cos(\pi t)$

$\pi t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$t = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$ seconds **[1 mark]**

e At rest, $v = 0$

$0 = -3\pi \sin(\pi t)$ **[1 mark]**

$0 = \sin(\pi t)$

$\pi t = 0, \pi, 2\pi, \ldots$

$t = 0, 1, 2, \ldots$ seconds **[1 mark]**

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When $t = 0$, $a = -3\pi^2 \cos(0) = -3\pi^2$

When $t = 1$, $a = -3\pi^2 \cos(\pi) = 3\pi^2$

When $t = 2$, $a = -3\pi^2 \cos(2\pi) = -3\pi^2$ and so on

Acceleration is $\pm 3\pi^2$ cm/s^2 **[2 marks]**

6 a When $t = 0$, $N = 2400e^0$

$= 2400$ people **[1 mark]**

b When $t = 6$, $N = 2400e^{-0.11 \times 6}$

$= 1240$ people **[1 mark]**

c $\dfrac{dN}{dt} = 2400 \times -0.11e^{-0.11t}$

$= -264e^{-0.11t}$ **[1 mark]**

i When $t = 0$, $\dfrac{dN}{dt} = -264e^{-0.11 \times 0}$

$= -264$ **[1 mark]**

Unemployment rate is decreasing by 264 people/year.

ii When $t = 6$, $\dfrac{dN}{dt} = -264e^{-0.11 \times 6}$

$= -136$ **[1 mark]**

Unemployment rate is decreasing by 136 people/year.

Part C – Analysis questions

1 a When $t = 0$, $V = 5000 + 1300e^{0.034 \times 0}$

$= 5000 + 1300$

$= 6300$ kL **[1 mark]**

b i When $t = 10$, $V = 5000 + 1300e^{0.034 \times 10}$

$= 6826.4$ kL **[1 mark]**

ii 6 months = 26 weeks

When $t = 26$, $V = 5000 + 1300e^{0.034 \times 26}$

$= 8146.7$ kL **[1 mark]**

c $\dfrac{dV}{dt} = 1300 \times 0.034e^{0.034t}$

$= 44.2e^{0.034t}$ **[1 mark]**

i When $t = 0$, $\dfrac{dV}{dt} = 44.2e^{0.034 \times 0}$

$= 44.2$ kL/week **[1 mark]**

ii When $t = 10$, $\dfrac{dV}{dt} = 44.2e^{0.034 \times 10}$

$= 62.1$ kL/week **[1 mark]**

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iii When $t = 26$, $\frac{dV}{dt} = 44.2e^{0.034 \times 26}$

$= 107$ kL/week **[1 mark]**

d $\frac{dV}{dt} = 0.034 \times 1300e^{0.034t}$

$= 0.034(5000 + 1300e^{0.034t} - 5000)$ **[1 mark]**

$= 0.034(V - 5000)$ **[1 mark]**

2 a Amplitude $a = 4.5$ **[1 mark]**

Period $= 2 \times 0.2$ (time to move to each side and back) **[1 mark]**

$\frac{2\pi}{b} = 0.4$

$\frac{2\pi}{0.4} = b$ **[1 mark]**

$5\pi = b$

$x = 4.5 \sin(5\pi t)$ **[1 mark]**

b $v = 4.5 \times 5\pi \cos(5\pi t)$

$= 22.5\pi \cos(5\pi t)$ **[1 mark]**

Amplitude $= 22.5\pi$, so the maximum speed is 22.5π cm/s. **[1 mark]**

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